Jungmin Kim

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WORK EXPERIENCE

Software Engineer (Machine Learning)

Vidrovr | September 2024 - Present

- Trained an animal classifier achieving 80.7% accuracy on iNaturalist dataset by implementing *semantic softmax loss* to leverage species taxonomies; tracked experiments in Comet and leveraged GPU acceleration for efficient training.
- Built and deployed a taxonomy-aware label expansion and filtering system using WordNet, enabling hierarchical suppression of tags (e.g., blocking "animal" also blocks "cat"), improving search and content filtering; integrated system-wide configuration via Kubernetes ConfigMaps for flexible production rollout.
- Added a new capability to the Re-Identification system for consolidating the identities of tracked objects within a video by enabling intra-video embedding comparisons.
- Redesigned the inference pipeline for NVIDIA NeMo models, converting them to ONNX as part of the process to remove framework-specific dependencies, resulting in improved stability in the production environment.
- Coordinated the step-by-step replacement of a database field, adding and backfilling the new field, retiring the old field, and aligning changes across frontend and backend systems
- Refactored model training codebase to PyTorch Lightning, reducing complexity and improving maintainability.
- Maintained >80% unit test coverage, ensuring reliable and production-ready code.

Software Engineering Intern (Machine Learning)

Vidrovr | June 2024 - September 2024

- Delivered a QR code recognition pipeline, including Postgres schema design, database migration scripts, new API endpoints, and API documentation.
- Researched compression effects in synthetic text generation to improve custom OCR training datasets.

ACADEMICS

Columbia University (2020~2024) | C. P. Davis Scholar

- B.S. in Computer Science, GPA: 3.83
- **Relevant Courseworks:** Machine Learning, First Principles of Computer Vision, Deep Learning in Computer Vision, Natural Language Processing, Machine Learning for Functional Genomics, Data Structures, Fundamentals of Computer Systems, Linear Algebra, Number Theory & Cryptography

Branksome Hall Asia (2013-2020) | International Baccalaureate Diploma | Graduated Top of the Class

- CEMC Fryer Contest (2017) - School Champion, Distinction, Individual Honor (Top 0.4%)

SKILLS

- **Languages:** Python, SQL, C, C++, Java, Swift, R, HTML&CSS, Javascript
- **Tools:** PyTorch, ONNX, NeMo, WordNet, Kubernetes, AWS, Docker, FastAPI, Postgres, Git

PERSONAL PROJECTS

iNAP: Nucleic Acid Binding Classification from a Protein Sequence Using BiLSTM-CNN *Jan 2023*

- Developed iNAP, a BiLSTM-CNN model using TensorFlow to classify proteins based on nucleic acid binding.
- Achieved an AUC of 0.922, outperforming existing models.
- Supports research in transcription, translation, splicing, apoptosis, and DNA repair.

NLP Model Implementations *Dec 2022*

- Trigram Model: Classified between human written text and GPT generated text with an **accuracy of 0.94**.
- Lexical Substitution: Built a context-aware synonym predictor using the Simple Lesk Algorithm and WordNet features such as synsets, hyponyms, hypernyms, definitions and examples.
- DependencyParserNN: Implemented Dependency Parsing algorithm and trained a Neural Network using Keras based on the parser's results to predict the dependency structure of an unlabeled sentence.
- Demonstrates integration of structured linguistic resources with NLP models.

Object Tracking and Highlight Reel Generation Pipeline *May 2024*

- Trained a YOLO model to detect surfers in a video.
- Developed a pipeline to generate cropped highlight videos based on tracked movements.
- Illustrates end-to-end computer vision system development from detection to automated output.

Gesture-based Real-time BlackJack Interface *May 2023*

- Implemented a real-time gesture recognition system using OpenCV to play Blackjack.
- Detected static and dynamic hand gestures ("hit," "check," "split," "double-down") with 0.84 recall.
- Showcases integration of computer vision with interactive real-time interfaces.